



B.Tech. Degree III Semester Supplementary Examination **May 2017**

ME 15-1302 ELECTRICAL TECHNOLOGY **(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain how the back emf of a motor causes the development of mechanical power.
- (b) Enumerate and classify the losses in a DC shunt motor.
- (c) State the condition which determines whether a dc machine is generating or motoring.
- (d) State and prove the condition for maximum efficiency of a transformer.
- (e) Draw the phasor diagram of a practical transformer on no load.
- (f) Explain the statement "All power in an autotransformer is not transformed inductively".
- (g) Define (i) pitch factor (ii) breadth factor.
- (h) A 6 pole induction motor is fed from 50 Hz supply. The frequency of rotor emf at full load is 2 Hz. Find the full load speed and slip.
- (i) Explain the effect of skin effect.
- (j) What are the advantages of high voltage transmission? State its limitations.

PART B

(4 × 10 = 40)

- II. (a) A 200 V shunt motor has $R_a = 0.1\Omega$, $R_f = 240\Omega$ and rotational loss 236W. On full load, the line current is 9.8A with the motor running at 1450 rpm. Determine (i) the mechanical power developed (ii) the power output (iii) the load torque (iv) the full load efficiency. (6)
 - (b) List the factors involved in voltage build up in a shunt generator. (4)
- OR**
- III. (a) Explain armature reaction with neat sketches. (5)
 - (b) At no load a dc shunt motor takes 5A from 400V supply. Armature and field resistance of the shunt motor is 0.5Ω , 200Ω . Find the efficiency of the motor when it takes 60 A line current. (5)
- IV. A 50 kVA, 2200/110 V, 50 Hz, 1 Φ transformer gave the following results: (10)
O.C. test : 400W, 10A, 110V
S.C. test : 808 W, 20.5A, 90V
(i) Draw the equivalent circuit referred to HV side.
(ii) Calculate the efficiency of the transformer at half load, 0.8 pf lagging.

OR**(P.T.O.)**

- V. (a) Derive EMF equation of transformer and show that emf/turn is same for both primary and secondary windings. (5)
- (b) A 500 kVA transformer has 95% efficiency at full load and also at 60% of full load both at upf. Separate out the losses. Determine the transformer efficiency at 75% full load, upf. (5)
- VI. (a) Why is synchronous motor not self starting? How can it be made self starting? (4)
- (b) A 3.3kV, 3 Φ star connected alternator has a full load current of 100A. Under short circuit it takes 5 A field current to produce full load short circuit current. The emf on open circuit for the same excitation is 900 V (line to line). The armature resistance is 0.9 Ω /phase. Determine the synchronous reactance per phase and voltage regulation for (i) 0.8 p.f. lagging (ii) 0.8 p.f. leading. (6)
- OR**
- VII. An 8pole, 3 phase, 50 Hz induction motor runs at a speed of 710 rpm with an input power of 35 kW. The stator copper loss at this condition is 1200 W while the rotational losses are 600W. Find (i) rotor Cu loss (ii) gross torque developed (iii) gross mechanical power developed (iv) net torque and mechanical power output. (10)
- VIII. With a neat schematic, explain the working of a hydro electric power plant. (10)
- OR**
- IX. (a) What are the electrical equipments in power station? Explain. (5)
- (b) Define corona. What are the factors affecting corona? (5)